DS 3010 Final Exam Practice

Spring 2026

Part 1 Conceptual Questions

Single Choice

Choose the best answer.

1. Which of the following statements about the bootstrap is **true**?
   (a) It requires multiple independent datasets.
   (b) It samples without replacement.
   (c) It samples with replacement from the original dataset.
   (d) It always reduces bias.

2. In KNN classification, as K increases:
   (a) Variance increases and bias decreases.
   (b) Variance decreases and bias increases.
   (c) Both bias and variance decrease.
   (d) Both bias and variance increase.

3. Which method assumes predictors are normally distributed within each class?
   (a) Logistic regression
   (b) LDA
   (c) KNN
   (d) ROC analysis

4. Which method assumes equal covariance matrices across classes?
   (a) Logistic regression
   (b) LDA
   (c) QDA
   (d) KNN

5. What does AUC measure?
   (a) The total training error of a classifier
   (b) The overall ability of a classifier to distinguish between classes across all thresholds
   (c) The number of predictors selected by the model
   (d) The variance of the predicted probabilities

6. What happens when we increase the classification threshold in logistic regression?
   (a) More observations are classified as $Y = 1$.
   (b) Fewer observations are classified as $Y = 1$.
   (c) The fitted coefficients change.
   (d) The training data changes.
7. What is the main problem with a single decision tree?
   (a) It is difficult to interpret.
   (b) It usually has high variance and can be unstable.
   (c) It cannot handle categorical predictors.
   (d) It always has high bias.
8. In bagging, what is the role of bootstrap sampling?
   (a) Reduce bias directly.
   (b) Create different training datasets.
   (c) Increase correlation between trees.
   (d) Select optimal splits.
9. In random forests, why do we use only $m < p$ predictors at each split?
   (a) Reduce bias.
   (b) Reduce correlation between trees.
   (c) Increase model interpretability.
   (d) Speed up prediction only.
10. What happens if $m = p$ in random forests?
    (a) The method becomes equivalent to bagging.
    (b) The trees become completely independent.
    (c) The model becomes a single decision tree.
    (d) The model can no longer use bootstrap samples.

Calculation Questions

11. Suppose you split a data set into a training set and a test set, both of which have a sample size of 2,500. For simplicity, let us assume that $p = 1$ only one predictor, and the response is $Y = 0$ or $Y = 1$. We obtain the following estimates for the training set:

$$\hat{\mu}_0 = 3.4, \qquad \hat{\mu}_1 = 5.1, \qquad \hat{\sigma}^2 = 4, \qquad \hat{\pi}_0 = 0.32, \qquad \hat{\pi}_1 = 0.68,$$

and for the test set:

$$\hat{\mu}_0 = 3.2, \qquad \hat{\mu}_1 = 5.5, \qquad \hat{\sigma}^2 = 4.5, \qquad \hat{\pi}_0 = 0.35, \qquad \hat{\pi}_1 = 0.65.$$

We assume that the predictor X_1 given the response follows a normal distribution and that the variance between the two classes is the same. Based on this information, write out the linear discriminant function for Y as a function of X_1. Explain when a test observation will be assigned to $Y = 0$ and when it will be assigned to $Y = 1$.

Hint: If you find yourself needing to do some complicated math derivation, then stop. Rethink the question. Your classification rule should involve real numbers from the estimates ($\hat{\mu}, \hat{\sigma}^2,$ etc.) given above.

12. Suppose we produce ten bootstrapped samples from a dataset containing red and green classes. We then apply a classification tree to each bootstrapped sample and, for a specific value of X, produce 10 estimates of $P(\text{class is red} \mid X)$:

$$0.1,\ 0.15,\ 0.2,\ 0.2,\ 0.55,\ 0.6,\ 0.6,\ 0.65,\ 0.7,\ 0.75.$$

There are two common ways to combine these results together in a single class prediction. One is the majority vote approach discussed in lecture. The second approach is to classify based on the average probability. In this example, what is the final classification under each of these approaches?

Part 2 Coding

Use the Hitters dataset to predict `Salary`. Remove any missing values. Log-transform the salaries and use that as your response. Create a training and test set.

1. Train a single regression tree. What is the training error? How many terminal nodes does the tree have?
2. Evaluate the tree on the test set. What is the test error?
3. Prune the tree. Which tree size corresponds to the lowest cross-validation error rate?
4. Report the training error and test error of the pruned tree.
5. Implement bagging ($m = p$). Fit the bagged model on the training set and evaluate it on the test set. Report its training error and test error.
6. Implement random forest (RF). Choose m in a data-driven way. Fit RF on the training set and evaluate it on the test set. Report its training error and test error.